

## Qualification Pack



# Industrial Control

QP Code: ASC/Q8319

Version: 1.0

NSQF Level: 4.5

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## Qualification Pack

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## Qualification Pack

### ASC/Q8319: Industrial Control

#### Brief Job Description

Industrial Control contains elements of both electrical installations and automation installations, with greater emphasis on automation installation. The industrial control practitioner requires a wide range of technical skills, such as installing conduits, cables, instruments, I/O devices and Programmable Logic Controllers. The industrial control practitioner also designs electrical circuits, programs Programmable Logic Controllers, parametrizes bus systems and configures Human Machine Interfaces.

#### Personal Attributes

The person should be result oriented with good technical and analytical skills, should have Excellent Interpersonal Skills, communication and presentation skills and a good team player. They should have ability to manage projects, prioritizing of work and mentoring the budding engineers.

#### Applicable National Occupational Standards (NOS)

##### Compulsory NOS:

1. [ASC/N9830: Work organization and management \(Industrial Control\)](#)
2. [ASC/N8368: Circuit design and modification](#)
3. [ASC/N8369: Making of automation control panels/centres](#)
4. [ASC/N8370: Field Installation \(electrical and automation\)](#)
5. [ASC/N8371: Programming \(Industrial control\)](#)
6. [ASC/N8372: Fault-finding](#)

#### Qualification Pack (QP) Parameters

|                   |                                |
|-------------------|--------------------------------|
| <b>Sector</b>     | Automotive                     |
| <b>Sub-Sector</b> | Manufacturing                  |
| <b>Occupation</b> | Automotive Product Development |
| <b>Country</b>    | India                          |
| <b>NSQF Level</b> | 4.5                            |

## Qualification Pack

|                                                           |                                |
|-----------------------------------------------------------|--------------------------------|
| <b>Credits</b>                                            | 17                             |
| <b>Aligned to NCO/ISCO/ISIC Code</b>                      | NCO-2015/2144.0801             |
| <b>Minimum Educational Qualification &amp; Experience</b> | No formal education prescribed |
| <b>Minimum Level of Education for Training in School</b>  |                                |
| <b>Pre-Requisite License or Training</b>                  | NA                             |
| <b>Minimum Job Entry Age</b>                              | 22 Years                       |
| <b>Last Reviewed On</b>                                   | NA                             |
| <b>Next Review Date</b>                                   | 08/02/2026                     |
| <b>NSQC Approval Date</b>                                 | 08/02/2024                     |
| <b>Version</b>                                            | 1.0                            |
| <b>Reference code on NQR</b>                              | QG-4.5-AU-01813-2024-V1-ASDC   |
| <b>NQR Version</b>                                        | 1                              |

## Qualification Pack

### ASC/N9830: Work organization and management (Industrial Control)

#### Description

This NOS unit is about implementing safety, planning work, adopting sustainable practices for optimising the use of resources.

#### Scope

The scope covers the following :

- Manage work and resources

#### Elements and Performance Criteria

##### *Manage work and resources*

To be competent, the user/individual on the job must be able to:

- PC1.** Consistently promote and comply with health and safety regulations and industry best practices in all working environments
- PC2.** Correctly use all safety equipment and personal protection equipment (PPE), lock off systems, and warning indicators
- PC3.** Recognize hazards and potentially hazardous situations and take appropriate actions to minimize risk to self and others
- PC4.** Work effectively as part of a team
- PC5.** Communicate effectively with other professionals including work-shop supervisors and other staff where installations are being carried out
- PC6.** Explain complex mechanical and engineering projects to colleagues who may not have specialist knowledge
- PC7.** Provide expert advice and guidance regarding on-going use, care, and maintenance of equipment
- PC8.** Think logically and work systematically
- PC9.** Reduce waste as much as possible and apply sorting

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Health and safety regulations and best practice, especially in relation to hazardous working environments and the variety of locations and industrial settings where the work may be conducted
- KU2.** Safety requirements relating to plant and equipment
- KU3.** SIL levels of safety and the application to relevant industries
- KU4.** The importance of site safety inductions
- KU5.** The range of safety equipment used to protect self and others and the application relating to various industries

## Qualification Pack

**KU6.** The types of hazards that may be encountered in industrial settings

**KU7.** The importance of effective communications and interpersonal skills

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                       | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Manage work and resources</i>                                                                                                                       | <b>15</b>    | <b>25</b>       | -             | <b>10</b>  |
| <b>PC1.</b> Consistently promote and comply with health and safety regulations and industry best practices in all working environments                 | 2            | 3               | -             | 1          |
| <b>PC2.</b> Correctly use all safety equipment and personal protection equipment (PPE), lock off systems, and warning indicators                       | 2            | 3               | -             | 2          |
| <b>PC3.</b> Recognize hazards and potentially hazardous situations and take appropriate actions to minimize risk to self and others                    | 2            | 3               | -             | 1          |
| <b>PC4.</b> Work effectively as part of a team                                                                                                         | 2            | 3               | -             | 1          |
| <b>PC5.</b> Communicate effectively with other professionals including work-shop supervisors and other staff where installations are being carried out | 2            | 4               | -             | 1          |
| <b>PC6.</b> Explain complex mechanical and engineering projects to colleagues who may not have specialist knowledge                                    | 1            | 2               | -             | 1          |
| <b>PC7.</b> Provide expert advice and guidance regarding on-going use, care, and maintenance of equipment                                              | 2            | 3               | -             | 1          |
| <b>PC8.</b> Think logically and work systematically                                                                                                    | 1            | 2               | -             | 1          |
| <b>PC9.</b> Reduce waste as much as possible and apply sorting                                                                                         | 1            | 2               | -             | 1          |
| <b>NOS Total</b>                                                                                                                                       | <b>15</b>    | <b>25</b>       | -             | <b>10</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                       |
|----------------------------|-------------------------------------------------------|
| <b>NOS Code</b>            | ASC/N9830                                             |
| <b>NOS Name</b>            | Work organization and management (Industrial Control) |
| <b>Sector</b>              | Automotive                                            |
| <b>Sub-Sector</b>          | Generic                                               |
| <b>Occupation</b>          | Generic                                               |
| <b>NSQF Level</b>          | 4.5                                                   |
| <b>Credits</b>             | 1                                                     |
| <b>Version</b>             | 1.0                                                   |
| <b>Last Reviewed Date</b>  | 08/02/2024                                            |
| <b>Next Review Date</b>    | 08/02/2026                                            |
| <b>NSQC Clearance Date</b> | 08/02/2024                                            |



## Qualification Pack

### ASC/N8368: Circuit design and modification

#### Description

This NOS is about preparing or modifying the circuit design of the system

#### Scope

The scope covers the following :

- Preparing or modifying the circuit design

#### Elements and Performance Criteria

##### *Preparing or modifying the circuit design*

To be competent, the user/individual on the job must be able to:

- PC1.** Read and interpret and make additions to technical diagrams in a simulation software according to a function description
- PC2.** Advise on modifications to circuit design
- PC3.** Interpret drawing standard sections (DIN ISO 1219) that are to be used
- PC4.** Design electrical circuits

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Principles of technical specification diagrams
- KU2.** Special technical terms and symbols
- KU3.** Principles and functions of relay/contactors circuits

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                     | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Preparing or modifying the circuit design</i>                                                                                     | <b>15</b>    | <b>25</b>       | -             | <b>10</b>  |
| <b>PC1.</b> Read and interpret and make additions to technical diagrams in a simulation software according to a function description | 3            | 5               | -             | 3          |
| <b>PC2.</b> Advise on modifications to circuit design                                                                                | 4            | 6               | -             | 2          |
| <b>PC3.</b> Interpret drawing standard sections (DIN ISO 1219) that are to be used                                                   | 4            | 7               | -             | 2          |
| <b>PC4.</b> Design electrical circuits                                                                                               | 4            | 7               | -             | 3          |
| <b>NOS Total</b>                                                                                                                     | <b>15</b>    | <b>25</b>       | -             | <b>10</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                 |
|----------------------------|---------------------------------|
| <b>NOS Code</b>            | ASC/N8368                       |
| <b>NOS Name</b>            | Circuit design and modification |
| <b>Sector</b>              | Automotive                      |
| <b>Sub-Sector</b>          | Manufacturing                   |
| <b>Occupation</b>          | Automotive Product Development  |
| <b>NSQF Level</b>          | 4.5                             |
| <b>Credits</b>             | 1                               |
| <b>Version</b>             | 1.0                             |
| <b>Last Reviewed Date</b>  | 08/02/2024                      |
| <b>Next Review Date</b>    | 08/02/2026                      |
| <b>NSQC Clearance Date</b> | 08/02/2024                      |

## Qualification Pack

### ASC/N8369: Making of automation control panels/centres

#### Description

This is about making automation control panels/centres of the system.

#### Scope

The scope covers the following :

- Making of automation control panels/centres

#### Elements and Performance Criteria

##### *Making of automation control panels/centres*

To be competent, the user/individual on the job must be able to:

- PC1.** Read, understand, and interpret complex technical drawing, circuit diagrams, layouts, function descriptions, and terminal drawings
- PC2.** Apply information from technical specifications to effective work planning and solutions to engineering and operational problems
- PC3.** Install ducts, terminals, components, and wiring of the control panel according to the drawings and given tolerances
- PC4.** Complete appropriate panel building operations according to specifications
- PC5.** Interpret operations manuals and follow guidelines and instructions

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Terminology and symbols used in technical specifications and diagrams
- KU2.** Principles of technical drawing, circuit diagrams, layouts, function descriptions, and terminal drawings
- KU3.** Uses and layout of operation manuals
- KU4.** Electrical and mechanical tools used in panel building activities, such as drilling and cutting
- KU5.** Lean manufacturing processes (wastes etc.)
- KU6.** Responsibility/liability to the customer (extra holes, dirt, damages)

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                               | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Making of automation control panels/centres</i>                                                                                             | <b>15</b>    | <b>25</b>       | -             | <b>10</b>  |
| <b>PC1.</b> Read, understand, and interpret complex technical drawing, circuit diagrams, layouts, function descriptions, and terminal drawings | 3            | 5               | -             | 2          |
| <b>PC2.</b> Apply information from technical specifications to effective work planning and solutions to engineering and operational problems   | 3            | 5               | -             | 2          |
| <b>PC3.</b> Install ducts, terminals, components, and wiring of the control panel according to the drawings and given tolerances               | 3            | 5               | -             | 2          |
| <b>PC4.</b> Complete appropriate panel building operations according to specifications                                                         | 3            | 5               | -             | 2          |
| <b>PC5.</b> Interpret operations manuals and follow guidelines and instructions                                                                | 3            | 5               | -             | 2          |
| <b>NOS Total</b>                                                                                                                               | <b>15</b>    | <b>25</b>       | -             | <b>10</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                             |
|----------------------------|---------------------------------------------|
| <b>NOS Code</b>            | ASC/N8369                                   |
| <b>NOS Name</b>            | Making of automation control panels/centres |
| <b>Sector</b>              | Automotive                                  |
| <b>Sub-Sector</b>          | Manufacturing                               |
| <b>Occupation</b>          | Automotive Product Development              |
| <b>NSQF Level</b>          | 4.5                                         |
| <b>Credits</b>             | 3                                           |
| <b>Version</b>             | 1.0                                         |
| <b>Last Reviewed Date</b>  | 08/02/2024                                  |
| <b>Next Review Date</b>    | 08/02/2026                                  |
| <b>NSQC Clearance Date</b> | 08/02/2024                                  |

## Qualification Pack

### ASC/N8370: Field Installation (electrical and automation)

#### Description

This NOS is about installation of electrical and automation components of the system in field.

#### Scope

The scope covers the following :

- Installation of electrical and automation components

#### Elements and Performance Criteria

##### *Installation of electrical and automation components*

To be competent, the user/individual on the job must be able to:

- PC1.** Measure and calculate the correct positions for the components to be installed
- PC2.** Prepare and install wire trays within given tolerances
- PC3.** Install conduits, cables, devices, instruments, and control centre fittings
- PC4.** Install complex cabling systems that combine power and communications
- PC5.** Plan work effectively to meet time schedule requirements
- PC6.** Use all tools effectively and safely without risk to self or others in the workplace
- PC7.** Test and commission installed equipment
- PC8.** Complete all necessary documentation following installation

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Issues and challenges of the installation of field components
- KU2.** Principles of technical drawings, layouts of installations and control panels, circuit diagrams, and flow charts
- KU3.** Principles and functions of all components used in field installation
- KU4.** Importance of accurate measuring and calculations during field installations

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Installation of electrical and automation components</i>                                      | <b>15</b>    | <b>25</b>       | -             | <b>10</b>  |
| <b>PC1.</b> Measure and calculate the correct positions for the components to be installed       | 2            | 3               | -             | 1          |
| <b>PC2.</b> Prepare and install wire trays within given tolerances                               | 2            | 3               | -             | 2          |
| <b>PC3.</b> Install conduits, cables, devices, instruments, and control centre fittings          | 2            | 3               | -             | 1          |
| <b>PC4.</b> Install complex cabling systems that combine power and communications                | 2            | 4               | -             | 2          |
| <b>PC5.</b> Plan work effectively to meet time schedule requirements                             | 2            | 3               | -             | 1          |
| <b>PC6.</b> Use all tools effectively and safely without risk to self or others in the workplace | 2            | 3               | -             | 1          |
| <b>PC7.</b> Test and commission installed equipment                                              | 2            | 3               | -             | 1          |
| <b>PC8.</b> Complete all necessary documentation following installation                          | 1            | 3               | -             | 1          |
| <b>NOS Total</b>                                                                                 | <b>15</b>    | <b>25</b>       | -             | <b>10</b>  |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                |
|----------------------------|------------------------------------------------|
| <b>NOS Code</b>            | ASC/N8370                                      |
| <b>NOS Name</b>            | Field Installation (electrical and automation) |
| <b>Sector</b>              | Automotive                                     |
| <b>Sub-Sector</b>          | Manufacturing                                  |
| <b>Occupation</b>          | Automotive Product Development                 |
| <b>NSQF Level</b>          | 4.5                                            |
| <b>Credits</b>             | 4                                              |
| <b>Version</b>             | 1.0                                            |
| <b>Last Reviewed Date</b>  | 08/02/2024                                     |
| <b>Next Review Date</b>    | 08/02/2026                                     |
| <b>NSQC Clearance Date</b> | 08/02/2024                                     |

## Qualification Pack

### ASC/N8371: Programming (Industrial control)

#### Description

This NOS is about carrying about programming of the installed system.

#### Scope

The scope covers the following :

- Programming of system

#### Elements and Performance Criteria

##### *Programming of system*

To be competent, the user/individual on the job must be able to:

- PC1.** Create programmes according to written specifications and diagrams
- PC2.** Configure the HMI-screens according to written specifications and diagrams
- PC3.** Configure HMI-screens to switch automatically to customised start screens
- PC4.** Configure the VSD or Servodrive as required in the function descriptions
- PC5.** Configure VSDs or Servodrives to startup without error messages
- PC6.** Test functions thoroughly and safely
- PC7.** Demonstrate functions to users and provide expert advice and guidance
- PC8.** Conform to IEC sequence programming specifications

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Principles of technical specifications and diagrams
- KU2.** Processes of controlling motors, valves, and other devices used in industrial control
- KU3.** HMI and PC-based HMI/Visualization to communicate with the PLC code
- KU4.** Setting of input limits
- KU5.** Uses of industry accepted equipment such as PLC, HMI, VFD/VSD, and distributed IO
- KU6.** Distributed IO based and industrial Bus Technologies
- KU7.** Industry 4.0 technology
- KU8.** IEC sequence-programming methods (IEC 61131-3)

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                       | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Programming of system</i>                                                           | <b>15</b>    | <b>25</b>       | -             | <b>10</b>  |
| <b>PC1.</b> Create programmes according to written specifications and diagrams         | 2            | 3               | -             | 1          |
| <b>PC2.</b> Configure the HMI-screens according to written specifications and diagrams | 2            | 3               | -             | 2          |
| <b>PC3.</b> Configure HMI-screens to switch automatically to customised start screens  | 2            | 3               | -             | 1          |
| <b>PC4.</b> Configure the VSD or Servodrive as required in the function descriptions   | 2            | 4               | -             | 2          |
| <b>PC5.</b> Configure VSDs or Servodrives to startup without error messages            | 2            | 3               | -             | 1          |
| <b>PC6.</b> Test functions thoroughly and safely                                       | 2            | 3               | -             | 1          |
| <b>PC7.</b> Demonstrate functions to users and provide expert advice and guidance      | 2            | 3               | -             | 1          |
| <b>PC8.</b> Conform to IEC sequence programming specifications                         | 1            | 3               | -             | 1          |
| <b>NOS Total</b>                                                                       | <b>15</b>    | <b>25</b>       | -             | <b>10</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                  |
|----------------------------|----------------------------------|
| <b>NOS Code</b>            | ASC/N8371                        |
| <b>NOS Name</b>            | Programming (Industrial control) |
| <b>Sector</b>              | Automotive                       |
| <b>Sub-Sector</b>          | Manufacturing                    |
| <b>Occupation</b>          | Automotive Product Development   |
| <b>NSQF Level</b>          | 4.5                              |
| <b>Credits</b>             | 6                                |
| <b>Version</b>             | 1.0                              |
| <b>Last Reviewed Date</b>  | 08/02/2024                       |
| <b>Next Review Date</b>    | 08/02/2026                       |
| <b>NSQC Clearance Date</b> | 08/02/2024                       |

## Qualification Pack

### ASC/N8372: Fault-finding

#### Description

This NOS is about diagnosing the system for identifying the faults in it.

#### Scope

The scope covers the following :

- Fault finding in system

#### Elements and Performance Criteria

##### *Fault finding in system*

To be competent, the user/individual on the job must be able to:

- PC1.** Apply all safety precautions
- PC2.** Read, understand, and interpret complex written specifications and diagrams, understanding all technical symbols
- PC3.** Analyse the correct principles of fault finding
- PC4.** Recognize incorrect principles of fault finding
- PC5.** Utilize the correct fault-finding principles
- PC6.** Utilize a range of tools and software to isolate faults

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Safety risks during the fault-finding process
- KU2.** Principles of written specifications, technical drawings, and circuit diagrams
- KU3.** Components and symbols of the relay-based circuit diagrams
- KU4.** Principles of the Relay Control Fault Finding using a multi-meter
- KU5.** Principles and functions of the common Industrial relay/contactors control circuits
- KU6.** Principles and functions of PLC diagnostics
- KU7.** Field Bus Diagnostic principles

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                             | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Fault finding in system</i>                                                                                               | <b>15</b>    | <b>25</b>       | -             | <b>10</b>  |
| <b>PC1.</b> Apply all safety precautions                                                                                     | 3            | 5               | -             | 2          |
| <b>PC2.</b> Read, understand, and interpret complex written specifications and diagrams, understanding all technical symbols | 3            | 5               | -             | 2          |
| <b>PC3.</b> Analyse the correct principles of fault finding                                                                  | 3            | 5               | -             | 2          |
| <b>PC4.</b> Recognize incorrect principles of fault finding                                                                  | 2            | 4               | -             | 2          |
| <b>PC5.</b> Utilize the correct fault-finding principles                                                                     | 2            | 3               | -             | 1          |
| <b>PC6.</b> Utilize a range of tools and software to isolate faults                                                          | 2            | 3               | -             | 1          |
| <b>NOS Total</b>                                                                                                             | <b>15</b>    | <b>25</b>       | -             | <b>10</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                |
|----------------------------|--------------------------------|
| <b>NOS Code</b>            | ASC/N8372                      |
| <b>NOS Name</b>            | Fault-finding                  |
| <b>Sector</b>              | Automotive                     |
| <b>Sub-Sector</b>          | Manufacturing                  |
| <b>Occupation</b>          | Automotive Product Development |
| <b>NSQF Level</b>          | 4.5                            |
| <b>Credits</b>             | 2                              |
| <b>Version</b>             | 1.0                            |
| <b>Last Reviewed Date</b>  | 08/02/2024                     |
| <b>Next Review Date</b>    | 08/02/2026                     |
| <b>NSQC Clearance Date</b> | 08/02/2024                     |

## Assessment Guidelines and Assessment Weightage

### Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training centre (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training centre based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.

## Qualification Pack

6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

### Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

## Assessment Weightage

Compulsory NOS

| National Occupational Standards                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage  |
|-----------------------------------------------------------------|--------------|-----------------|---------------|------------|-------------|------------|
| ASC/N9830.Work organization and management (Industrial Control) | 15           | 25              | 0             | 10         | 50          | 10         |
| ASC/N8368.Circuit design and modification                       | 15           | 25              | 0             | 10         | 50          | 8          |
| ASC/N8369.Making of automation control panels/centres           | 15           | 25              | 0             | 10         | 50          | 15         |
| ASC/N8370.Field Installation (electrical and automation)        | 15           | 25              | 0             | 10         | 50          | 25         |
| ASC/N8371.Programming (Industrial control)                      | 15           | 25              | 0             | 10         | 50          | 34         |
| ASC/N8372.Fault-finding                                         | 15           | 25              | 0             | 10         | 50          | 8          |
| <b>Total</b>                                                    | <b>90</b>    | <b>150</b>      | <b>0</b>      | <b>60</b>  | <b>300</b>  | <b>100</b> |



## Qualification Pack

### Acronyms

|             |                                                 |
|-------------|-------------------------------------------------|
| <b>NOS</b>  | National Occupational Standard(s)               |
| <b>NSQF</b> | National Skills Qualifications Framework        |
| <b>QP</b>   | Qualifications Pack                             |
| <b>TVET</b> | Technical and Vocational Education and Training |

## Qualification Pack

### Glossary

|                                              |                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sector</b>                                | Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.                                                                      |
| <b>Sub-sector</b>                            | Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.                                                                                                                                                                                     |
| <b>Occupation</b>                            | Occupation is a set of job roles, which perform similar/ related set of functions in an industry.                                                                                                                                                                                                |
| <b>Job role</b>                              | Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.                                                                                                                                                                                |
| <b>Occupational Standards (OS)</b>           | OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts. |
| <b>Performance Criteria (PC)</b>             | Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.                                                                                                                                                                    |
| <b>National Occupational Standards (NOS)</b> | NOS are occupational standards which apply uniquely in the Indian context.                                                                                                                                                                                                                       |
| <b>Qualifications Pack (QP)</b>              | QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.                                                                                                                       |
| <b>Unit Code</b>                             | Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'                                                                                                                                                                                                        |
| <b>Unit Title</b>                            | Unit title gives a clear overall statement about what the incumbent should be able to do.                                                                                                                                                                                                        |
| <b>Description</b>                           | Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.                                                                                                                   |
| <b>Scope</b>                                 | Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.                                                                                            |

## Qualification Pack

|                                         |                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge and Understanding (KU)</b> | Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.                                                                                                                                               |
| <b>Organisational Context</b>           | Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.                                                                                                                                                                       |
| <b>Technical Knowledge</b>              | Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.                                                                                                                                                                                                                                                               |
| <b>Core Skills/ Generic Skills (GS)</b> | Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles. |
| <b>Electives</b>                        | Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.                                                                            |
| <b>Options</b>                          | Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.                                                                                                                                                          |